



FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 31.07.2026

In addition to part I (General Handout for all courses appended to the Timetable) this portion gives further specific details regarding the course.

Course No : CE F231

Course Title : Fluid Mechanics

Instructor-in-Charge : Dr. Guntu Ravikumar (M W 5 Th 10 at 6104)

Tutorial Instructors : Dr. Guntu Ravikumar, Mr. Utkarsh Chaudhry, Ms. N Archana (F 1: T1, T2 at 6105)

1. Course Description:

Fluid Mechanics is an introductory course that covers the fundamental aspects of fluid mechanics, especially the classical and well-established laws. This course emphasizes understanding these governing laws and their widespread engineering applications in fluid related problems by covering the following topics: Concepts and definitions, compressibility of fluids, fluid pressure and measurement, fluid statics, buoyancy, rigid body motion, fluid kinematics, conservation laws: control volume approach, differential analysis of fluid flow, study of flow pattern through orifices and mouthpieces, notches and weirs, analysis of flow through pipes, viscous fluid flow analysis: analysis of flow through pipes, dimensional analysis and similitude.

The contents of this course will enable the students to tackle real-life fluid mechanics problems and provide a broad overview of the subject. Moreover, this course will be helpful to students as a prerequisite for further advanced courses in fluid mechanics at undergraduate and graduate levels.

2. Scope and Objective of the Course:

- Introduction to different fluid properties and their engineering applications
- Learn about pressure exerted by fluids and pressure measurement techniques
- Estimate hydrostatic forces on surfaces
- Learn about buoyancy and evaluate stability of submerged and floating bodies
- Understand concepts of fluid kinematics, methods of describing fluid flow, types of fluid flow, control mass and control volume, rate of flow, continuity equation, potential and stream functions
- Study dynamics of fluid flow, Euler's equation, Bernoulli's equation and its applications, Navier Stokes equation, and pipe bend problems
- Compute fluid discharge through orifices and mouthpieces, and flow over weirs and notches
- Analyze flow of viscous fluids through pipes and plates, and compute head loss due to friction
- Compute different types of energy losses and determine hydraulic gradient and total energy line for flow through pipes. Become adept at solving problems pertaining to syphon, flow through pipes in series, parallel and branched pipes, power transmission through pipes, and flow through nozzles
- Perceive the concepts of dimensional analysis and similitude

3. Course Learning Outcomes:

On completion of this course, the students will be able to,

1. Demonstrate the difference between solid and fluid, its properties and properties in static and dynamic conditions
2. Estimate pressure exerted by fluids and experiment with pressure measurement devices



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3. Apply conservation laws applicable to fluids and its applications through fluid kinematics and dynamics
4. Perceive and formulate relationship among the parameters involved in the given fluid phenomenon and apply concepts of similitude in analyzing models and prototypes
5. Estimate the losses in pipelines for both laminar and turbulent conditions and analysis of pipes connected in series and parallel.

4. Textbook:

TB1: Munson, B.R., Okiishi, T.H., Huebsch, W.W., Rothmayer, A.P., 2009. *Fundamentals of fluid mechanics*, 6th edition, Wiley India Pvt. Ltd, New Delhi.

TB2: Modi, M. and Seth, H., *Fluid mechanics and hydraulics*. Standard Book House, New Delhi.

5. Reference Book:

RB1: Som, S.K. and Biswas, G., 2004. *Introduction to Fluid Mechanics and Fluid Machines*, 2nd edition, Tata McGraw Hill Education, Noida, India.

RB2: Cimbala, J. and Cengel, Y., 2019. *Fluid Mechanics: Fundamentals and applications (in SI units)*, 4th edition, Tata McGraw Hill Education, New Delhi, India.

RB3: Fox, R.W. and McDonald, A.T., *Introduction to Fluid Mechanics*, John Wiley & Sons Inc., Singapore, Seventh Edition, 2011.

6. Course Plan:

Module No.	Lecture Session *	Reference	Learning outcomes
1 Week 1	Introduction and Properties of fluids: L1-L3	TB1 – Ch 1 TB2 – Ch 1	Learn about different fluid properties and their engineering applications
2 Week 2	Fluid pressure and its measurement: L4-L5	TB1 – Ch 2, TB2 – Ch 2	Learn about pressure exerted by fluids and pressure measurement techniques
3 Week 3	Hydrostatic forces on surfaces: L6-L9	TB1 – Ch 2, TB2 – Ch 3	Procedure to estimate hydrostatic forces on surfaces
4 Week 4	Buoyancy and floatation: L10-L12	TB1 – Ch 2, TB2 – Ch 4	Learn about buoyancy and evaluate stability of submerged and floating bodies
5 Week 5 and 6	Kinematics of flow: L13-L19	TB1 – Ch 3, Ch 4 TB2 – Ch 5, Ch 6	Understand concepts of fluid kinematics, methods of describing fluid flow, types of fluid flow, control mass and control volume, Reynolds transport theorem, continuity equation, potential and stream functions



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6 Week 7 and 8	Dynamics of fluid flow: L20-L25	TB1 – Ch 5, Ch 6, Ch 7 TB2 – Ch 7, Ch 8	Learn about dynamics of fluid flow, Euler's equation, Bernoulli's equation and its applications, Navier Stokes equation, and pipe bend problems
7 Week 8 and 9	Flow measurement: Orifices, mouthpieces, notches, and weirs: L26-L30	TB1 – Ch 8, TB2 – Ch 9, Ch 10	Compute fluid discharge through orifices, mouthpieces, notches, and weirs
8 Week 9 and 10	Viscous flow: L31-L33	TB1 – Ch 8, TB2 – Ch 11	Analyze flow of viscous fluids through pipes and plates, and compute head loss due to friction in viscous flow
9 Week 11 and 12	Flow through pipes: L34-L36	TB1 – Ch 8 TB2 – Ch 12, Ch 13, Ch 14, Lecture PPT	Compute different types of energy losses and determine hydraulic gradient and total energy line for flow through pipes. Become adept at solving problems pertaining to syphon, flow through pipes in series, parallel and branched pipes, power transmission through pipes, and flow through nozzles
10 Week 13	Dimensional and model analyses: L37-L38	TB1 – Ch 7 TB2 – Ch 17	Perceive and apply the concepts of dimensional analysis and similitude

7. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Class discussions (Linked to attendance)	5-10 mins.	10%	Every class	-
Evaluative Tutorials	45 mins.	30%	Every week	Close Book
Mid-Semester Test	90 mins.	25%	07/10 AN1	Close Book/Open Book
Comprehensive Examination	180 mins.	35%	10/12 FN	Close Book/Open Book



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- 8. Chamber Consultation Hour:** Friday 9AM – 10 AM. Chamber location for meeting: 1210-F.
- 9. Notices:** All announcements will be communicated via Nalanda (LMS Portal) and email.
- 10. Make-up Policy:** No make-up exams will be conducted for quizzes and Evaluative Tutorials. Make-up request for examination must be notified before the examination and decision will be made as per institute norms. Make-up requests made on medical grounds may not be approved unless recommended by CMO BITS Pilani.

Instructor-in-charge
CE F231